

# ARTHUR ARIES

Mulhouse, France | +33 6 80 52 49 81 | arthur.aries@outlook.com  
arthuraries.com | linkedin.com/in/arthur-aries

## Computer Vision & Embedded AI Engineer

French AI Engineer (6+ years exp) specializing in Computer Vision and Embedded Deep Learning. Expert in bridging the gap between advanced ML research and robust software engineering.

**Target Role:** Seeking challenging positions as a **Computer Vision / Embedded AI Engineer** (open to broader AI/ML), focusing on high-performance edge applications.

### PROFESSIONAL EXPERIENCE

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#### Consultant – AI & Computer Vision Engineer

Nov. 2020 – Present

*Technology & Strategy, Strasbourg (FR) & Stuttgart (DE)*

- **Noremat (Autonomous Mowing)**

- Engineered embedded semantic segmentation (DeepLabV3+, Jetson Orin NX).
- Optimized pipeline (ROS 2/TensorRT) for 10 FPS (<100ms latency) with safety-critical fallback behaviors.

- **ARQUUS Defense (Path Detection)**

- Delivered off-road navigation demonstrator; refactored training pipeline across 15+ architectures.
- Orchestrated continuous CI-benchmarking to ensure robust validation in diverse environments.

- **AimFlow (MLOps Platform - Sole Engineer)**

- Conceived and built a standalone MLOps desktop application (Tauri/React) to automate the full model lifecycle (MLflow, Optuna).

- **AimLab (Annotation - End-to-End Ownership)**

- Delivered production-ready tool integrating SAM 2 for zero-shot assisted annotation, significantly accelerating R&D iteration loops.

- **Bosch (Real-Time Edge AI)**

- Deployed real-time object detection (YOLO/NanoDet) on NXP i.MX 8 hardware.
- Implemented pseudo-labeling algorithms slashing manual annotation overhead by 50-80%.

### PERSONAL PROJECTS

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- **BandejAI (Padel Tactics Engine):** Engineered an offline analysis pipeline utilizing YOLOv8 tracking and 2D court homography, laying the groundwork for a multimodal Virtual Coach.
- **FormulIA1 (F1 CV Pipeline):** Developed an end-to-end real-time F1 video analysis pipeline featuring YOLO object detection, occlusion-aware ReID tracking, and dynamic HUD overlays.

### TECHNICAL SKILLS & LANGUAGES

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- **Languages:** Python, C, C++, TypeScript.
- **AI & Computer Vision:** PyTorch, TensorFlow, OpenCV, SAM 2, DeepLabV3+, YOLO, SMP.
- **Embedded & Deployment:** NVIDIA Jetson (Orin NX), TensorRT, ONNX, ROS 2, Isaac ROS, GStreamer.
- **Tools & MLOps:** Linux, Docker, MLflow, Optuna, FastAPI, React, Tauri, SQLite.
- **Languages:** French (Native), English (Fluent - C1), Chinese (Notions), Spanish (Notions).

### EDUCATION, CERTIFICATIONS & INTERESTS

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**Education:** EISTI - Master in CS/AI (2020) | EPF - Integrated Scientific Prep. Program (2016).

**Certifications:** IBM Data Science & Applied AI, UC Irvine IoT/Embedded Systems (*Coursera*).

**Interests:** Padel, Football, F1, Esports, Game Development, Algorithmic Strategy, Reading.